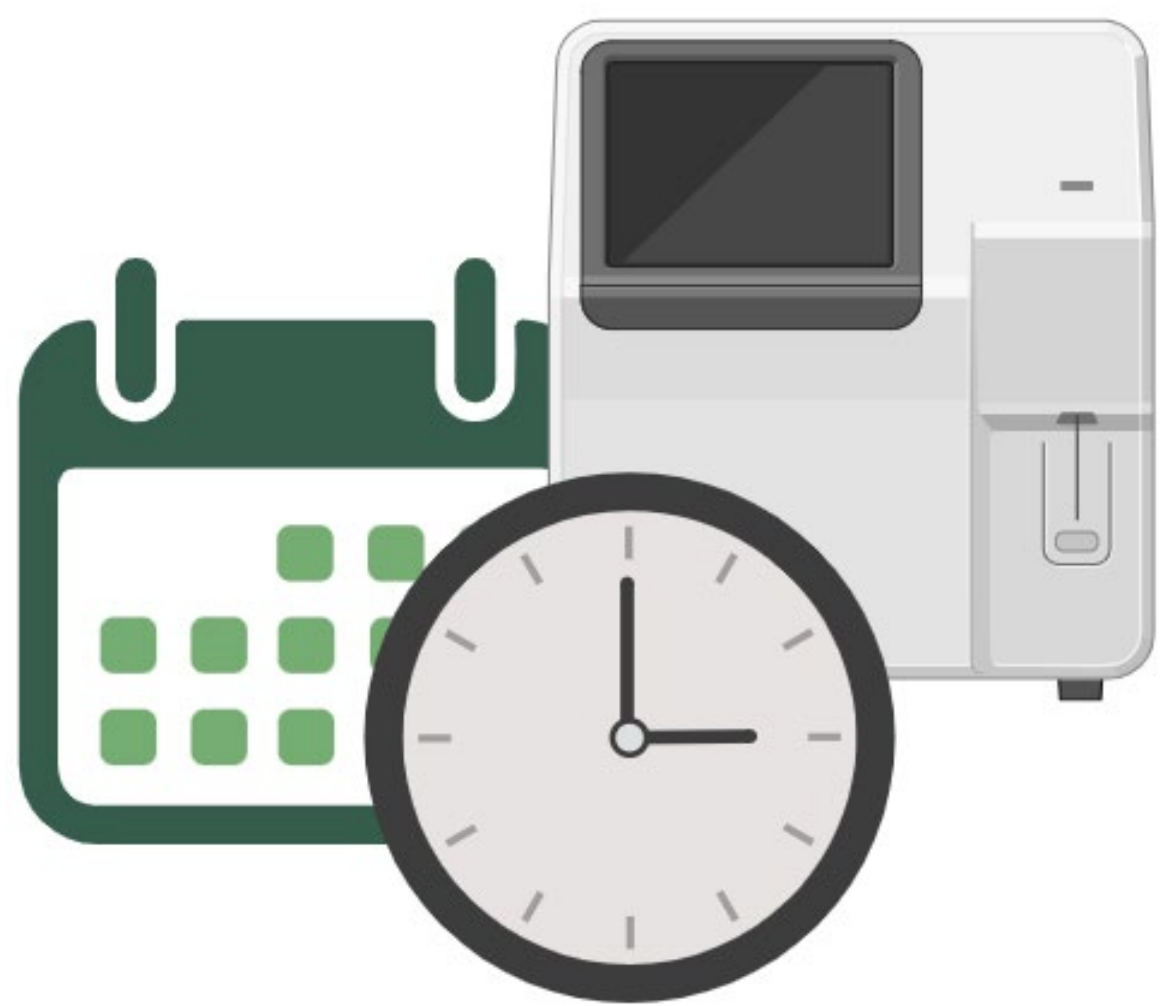
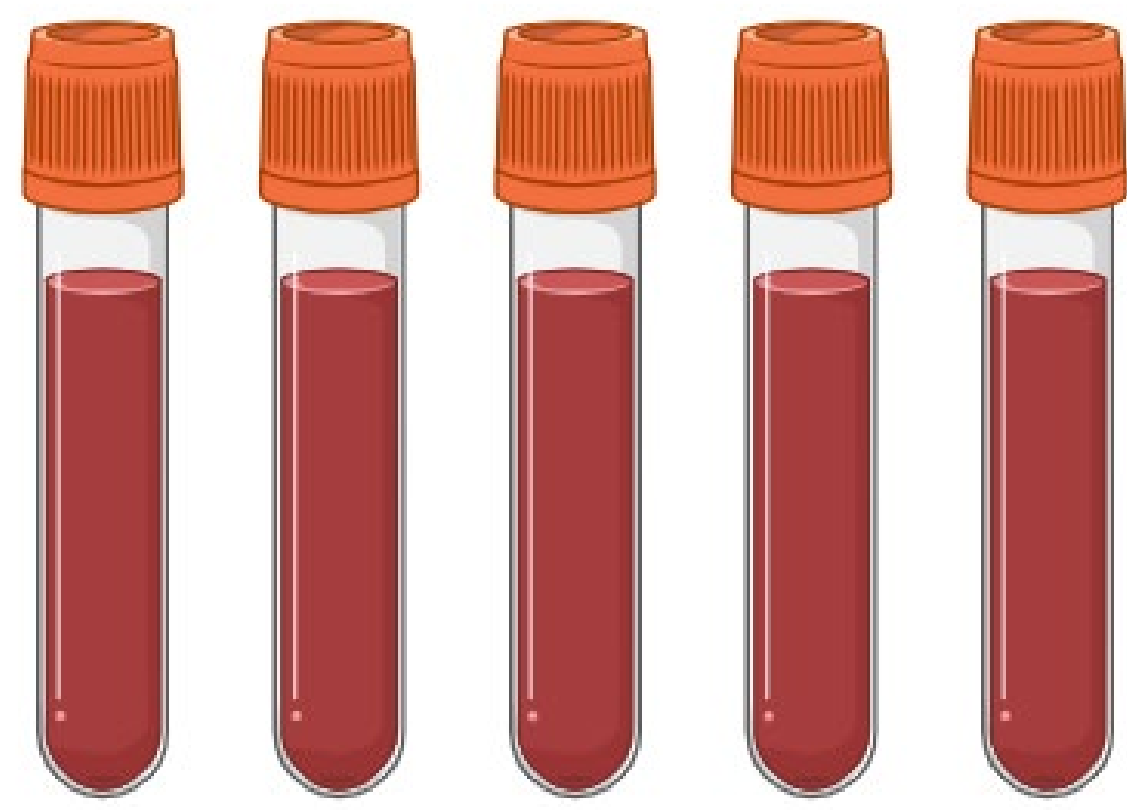


HOW DO WE TEST THE STABILITY OF FRESH BLOOD SAMPLES?

Blood Count and Flow Cytometry EQA Programmes



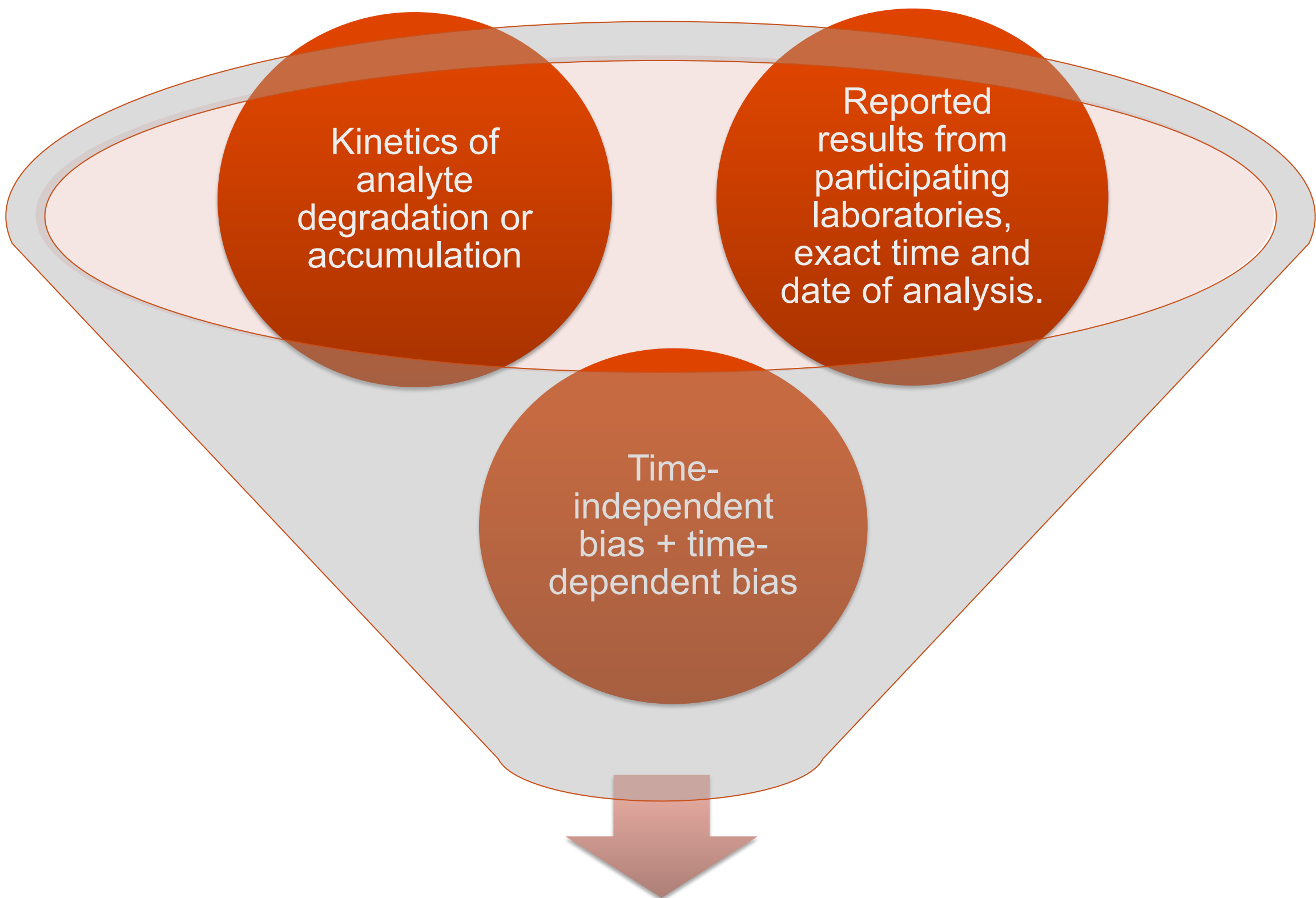
Help us improve EQA accuracy
Precise reporting of the date and time of analysis is essential for a reliable assessment of stability.

WHY STABILITY MATTERS IN EQA ?

In EQA, laboratories are evaluated based on how close their results are to a target value.

If a sample is unstable, its analyte concentration may change over time, introducing bias that affects laboratory performance scores (z-scores and u-scores).

For haematology analyses, this bias is often time-dependent, making it crucial to understand how analyte levels evolve during the testing period.



Stability is considered sufficient if the time-related changes do not significantly affect laboratory evaluation outcomes.

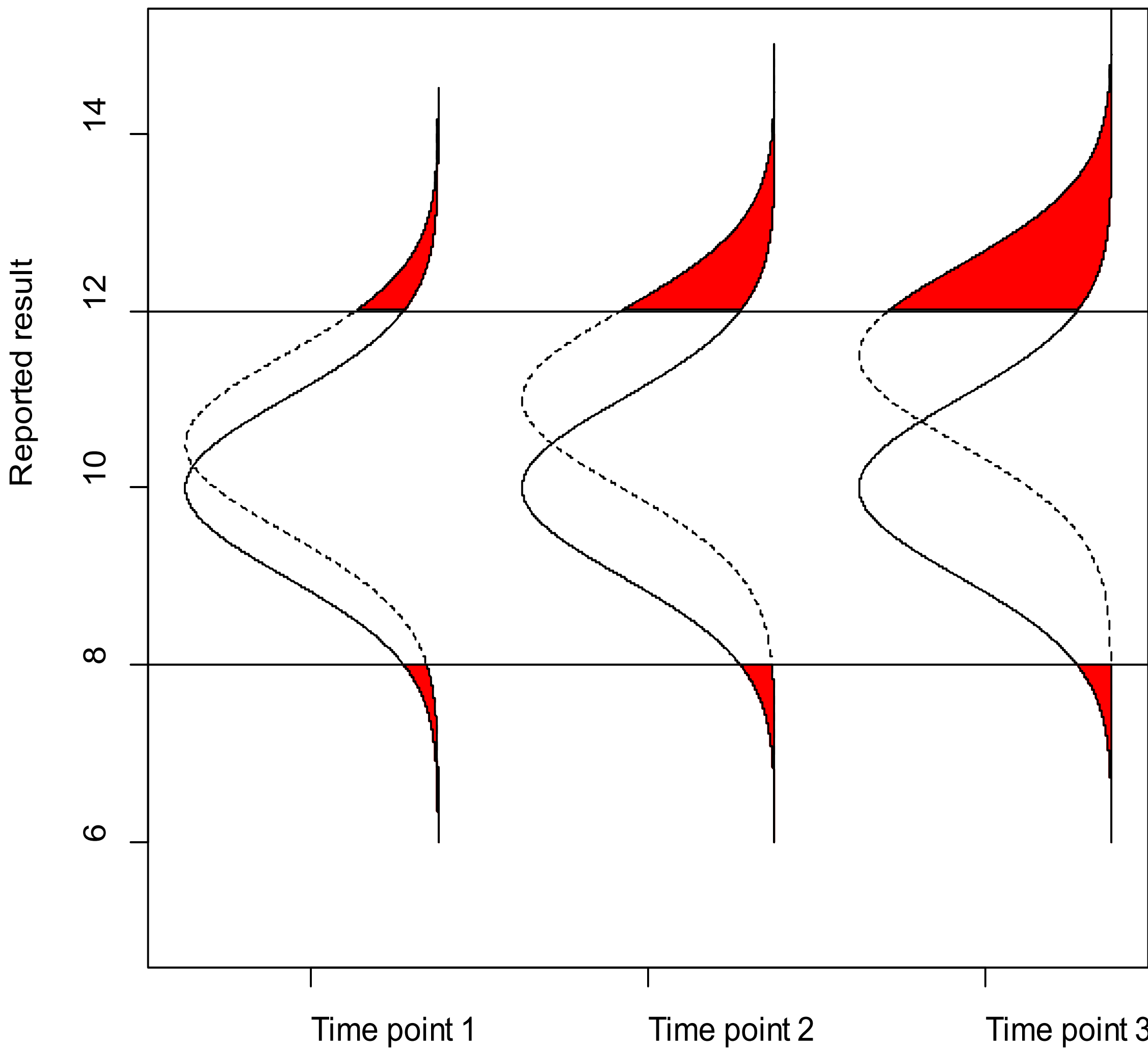


Figure 1: How instability affects EQA data? Solid lines represent the expected data distribution under stable conditions, while dashed lines show the distribution under instability, indicated by a gradual shift. Red areas highlight the differences in the likelihood of flagging results between stable and unstable scenarios.

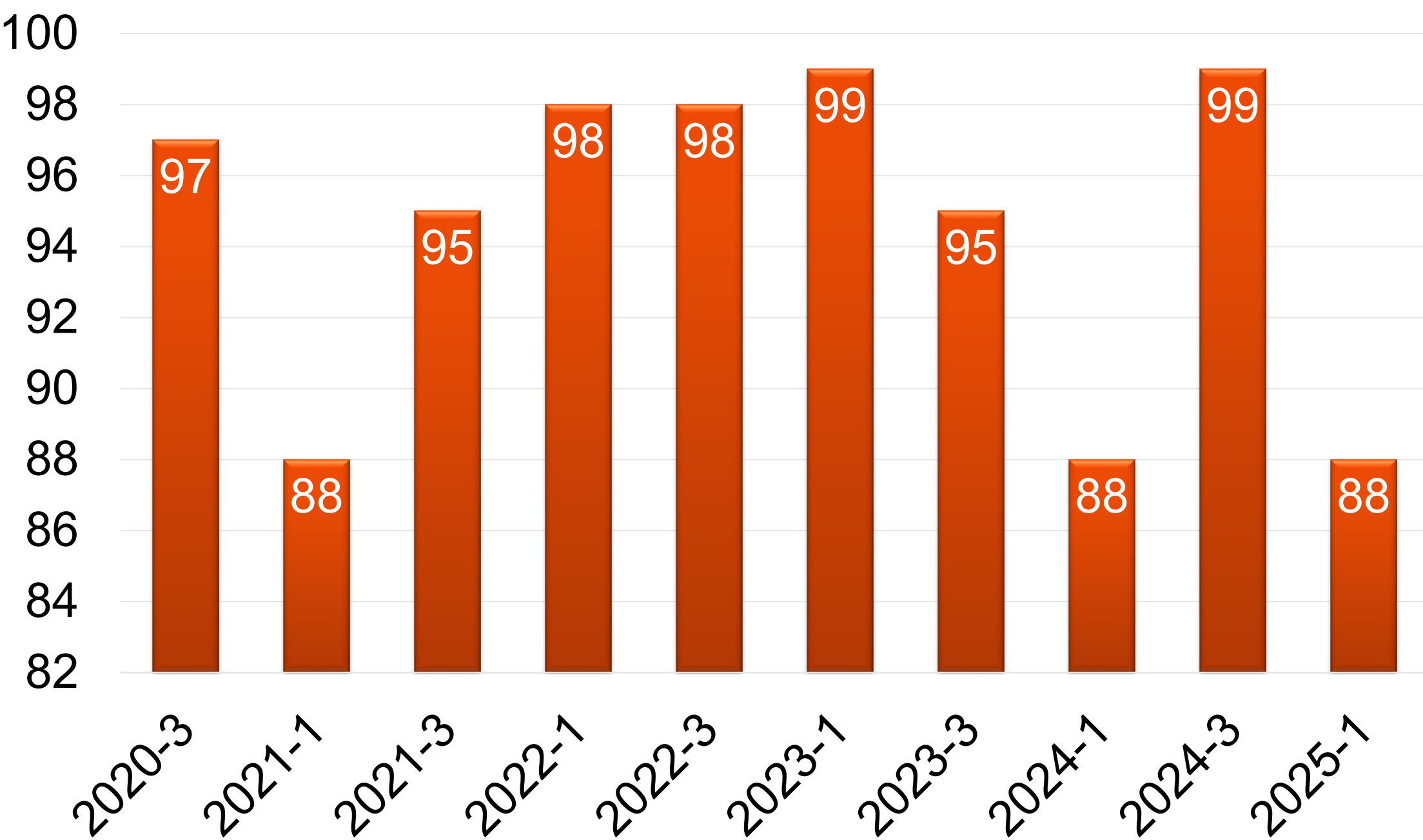


Figure 2: Percentage of laboratories that completed analysis within 48 hours over the past five years (Blood Count Program).

OUR APPROACH

By integrating time of analysis into the evaluation process, we gain a powerful tool to assess sample stability in real-world conditions. This method is efficient, cost-effective, and aligns with the practical realities of EQA programs.

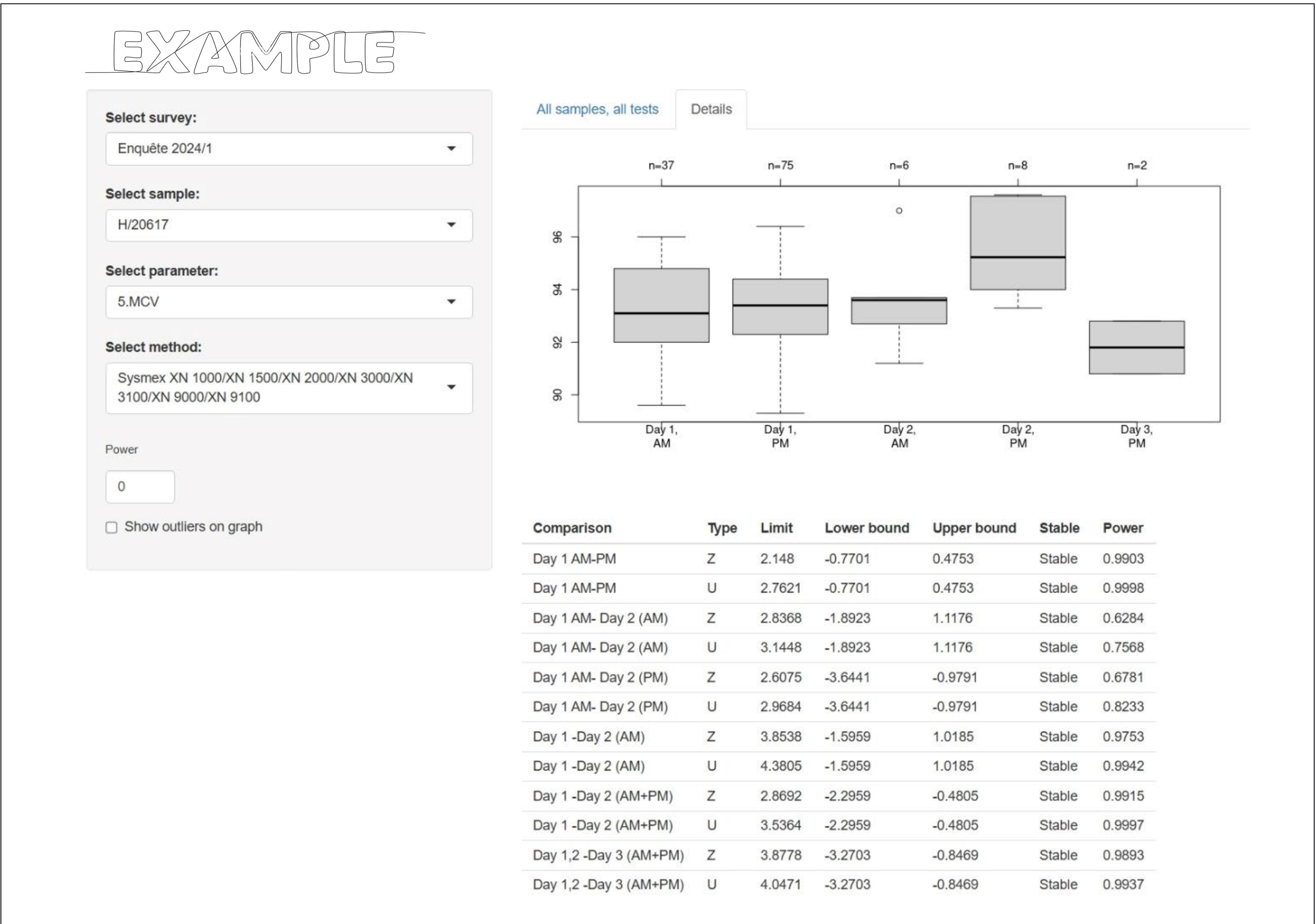


Figure 3: Illustrative example from our stability testing application.